

Project: Mission-Critical Incident Management System (IMS)

Summary

This submission implements a resilient Incident Management System for high-volume distributed-stack signals across APIs, MCP hosts, caches, queues, RDBMS, and NoSQL stores. The system ingests signals asynchronously, applies API-key security and rate limiting, debounces noisy component failures, persists raw and structured data separately, exposes operational metrics, and provides a workflow-driven React dashboard with mandatory RCA before closure.

How to run

1. `IMS_API_KEY=dev-secret docker compose -f docker-compose.production.yml up --build`
2. Open `http://localhost:5173` for the dashboard.
3. Open `http://localhost:8000/health` for health.
4. Open `http://localhost:9090/query` for Prometheus.
5. Seed data: `cd backend && python3 scripts/simulate_failure.py --base-url http://localhost:8000 --api-key dev-secret --rate 1000 --duration 1 --batch-size 500`

Rubric coverage

- Concurrency and scaling: FastAPI, asyncio workers, bounded queue of 25,000, batch flushing at 1,000 signals or 1 second, immediate 503 backpressure.
- Data handling: JSONL raw signal audit log, SQLite source of truth, Redis-backed production hot cache and aggregation profile, in-memory local fallback.
- LLD: Alerting Strategy pattern and Workflow State pattern.
- UI/UX: React/Vite dashboard with live feed, detail view, raw signals, status workflow, RCA form, and audit timeline.
- Resilience: exponential retry decorator for SQLite writes, graceful shutdown drain, health checks, Prometheus metrics, signal sanitization.
- Testing: pytest suite covers alerting, RCA validation, batch processing, backpressure, sanitization, security, and health/metrics.
- Documentation: README, benchmark report, production upgrade path, rubric mapping, screenshots guide, prompt/context/plan markdowns.

MTTR formula

$$\text{MTTR} = \frac{\text{sum(Resolution Timestamp - Incident Start Timestamp)}}{\text{Total Incidents Resolved}}$$

Verification

Backend tests: 13 passed. Frontend production build: successful.